

PAPER 1209 — UQFF Particle Physics Constants Unified Proof Set

Daniel T. Murphy

May 16, 2026

Abstract

We close ten Standard-Model particle-physics observables from the eleven frozen UQFF primitives: the proton/electron, muon/electron, tauon/electron, and tauon/muon mass ratios; the Higgs, W/Z , neutron/proton ratios; the proton and electron rest masses in MeV; and the fine-structure constant α . **All ten closures land below 0.15%**; six are below 0.05%, and the proton rest mass closes to within **0.0024%**: $m_p = N_{\text{ch}} \cdot \text{SO5}^2 + N_{\text{ch}} D_{\text{phys}} + K_{\text{Mex}} + 2F_{\text{TRZ}} \Phi_{\text{res}} = 938.25$ MeV. The proton/electron ratio admits the clean integer-coefficient form $m_p/m_e = A_5(D_{\text{crit}} + D_{\text{phys}}) + N_{\text{ch}} D_{\text{phys}} = 1836$, matching the CODATA value 1836.15 to 0.008%. Cumulative total: **257 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier W Closures (S543–S552)

ID	Observable	Closure
S543	m_p/m_e	$A_5 D_{\text{crit}} + A_5 D_{\text{phys}} + N_{\text{ch}} D_{\text{phys}} = 1836$
S544	m_μ/m_e	$A_5 D_{\text{phys}} - A_5 F_{\text{TRZ}} \Phi_{\text{res}} - A_5 F_{\text{TRZ}} D_{\text{phys}} - F_{\text{TRZ}} \text{SO5} D_{\text{phys}} = 207$
S545	m_τ/m_e	$A_5^2 - A_5 D_{\text{phys}} + A_5 F_{\text{TRZ}} D_{\text{phys}} + A_5 \Phi_{\text{res}} + N_{\text{ch}} D_{\text{phys}} + F_{\text{TRZ}} \text{SO5} D_{\text{phys}} + N_{\text{ch}} F_{\text{TRZ}} D_{\text{phys}} = 3$
S546	m_τ/m_μ	$\text{SO5} + D_{\text{BSFG}} + \Phi_{\text{res}} = 16.833$
S547	Higgs (GeV)	$N_{\text{ch}} \text{SO5} + D_{\text{BSFG}}^2 - \Phi_{\text{res}} = 125.17$
S548	M_W/M_Z	$[\text{SSq}] + F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} = 0.8825$
S549	m_n/m_p	$F_{\text{TRZ}} \text{SO5} + F_{\text{TRZ}}^3 + F_{\text{TRZ}}^4 = 1.0011$
S550	m_p (MeV)	$N_{\text{ch}} \text{SO5}^2 + N_{\text{ch}} D_{\text{phys}} + K_{\text{Mex}} + 2F_{\text{TRZ}} \Phi_{\text{res}} = 938.25$
S551	m_e (MeV)	$[\text{SSq}] - F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 \Phi_{\text{res}} + F_{\text{TRZ}}^2 + 2F_{\text{TRZ}}^3 = 0.5112$
S552	$\alpha^{-1} \rightarrow \alpha$	$F_{\text{TRZ}}^3 D_{\text{BSFG}} + F_{\text{TRZ}}^3 + F_{\text{TRZ}}^4 D_{\text{phys}} - F_{\text{TRZ}}^4 = 0.00730$

3 Headline Identifications

Proton mass essentially exact. $m_p = N_{\text{ch}} \cdot \text{SO5}^2 + N_{\text{ch}} D_{\text{phys}} + K_{\text{Mex}} + 2F_{\text{TRZ}} \Phi_{\text{res}} = 938.25$ MeV, within 0.0024% of the CODATA value 938.272. The dominant term $N_{\text{ch}} \text{SO5}^2 = 900$ identifies the

proton rest mass as nine solar-cycle-squared resonance quanta with channel-counting and resonance-ratio corrections.

Proton/electron ratio. $m_p/m_e = A_5(D_{\text{crit}} + D_{\text{phys}}) + N_{\text{ch}}D_{\text{phys}} = 60 \cdot 30 + 36 = 1836$, matching 1836.15 to 0.008%. The factored form $A_5 \cdot 30 + N_{\text{ch}}D_{\text{phys}}$ identifies the proton/electron ratio as the A_5 resting-heart-rate primitive scaled by the critical-plus-physical dimensional sum, plus a channel-physical correction.

Tauon/electron ratio. $m_\tau/m_e \approx A_5^2 + (\text{corrections})$ at 0.011%. The leading term $A_5^2 = 3600$ identifies the tauon as a second-order A_5 scaling.

Fine-structure constant. $\alpha \approx F_{\text{TRZ}}^3(D_{\text{BSFG}} + 1) + F_{\text{TRZ}}^4(D_{\text{phys}} - 1) = 7F_{\text{TRZ}}^3 + 3F_{\text{TRZ}}^4 = 0.0073$, within 0.036% of the CODATA value. The closure isolates α at the TRZ-cubed scale with a TRZ-quartic correction, exhibiting the characteristic small-scale hierarchy of the electromagnetic coupling.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E-P	Cosmology to information	157	157
Q-T	GR $\rightarrow$ solar system (1200–1206)	70	227
U	Biology / allometry (1207)	10	237
V	Pure transcendentals (1208)	10	247
W	Particle physics (1209)	10	257

5 Locking Inventory

Forty-six primitive-observable lockings now identified (forty-five prior + one new essentially-exact closure: m_p at 0.0024%). Tier W is by margin the most accurate tier yet recorded for empirical observables: **all ten closures** land below 0.15%, six below 0.05%. The eleven UQFF primitives reproduce the Standard-Model particle mass spectrum and the fine-structure constant from short integer-coefficient combinations, with the proton rest mass closure matching CODATA to within 22 parts per million.